

Agentic AI for DNS-Based Threat Intelligence

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Abstract

This project presents an agentic, DNS-based threat intelligence pipeline centered on automated enrichment, with infrastructure clustering and similarity-driven threat actor attribution implemented as complementary analytical components. The system leverages DNS and registration artifacts to infer relationships between domains and support campaign-level analysis under weak supervision.

The pipeline is implemented as a modular architecture orchestrated through n8n for automated ingestion and enrichment, while clustering and attribution are a complementary offline analytical subsystem implemented in Python, with future work focused on integration of clustering and attribution into the live n8n workflow and incorporation of additional behavioral signals.

Experimental evaluation on a mixed dataset of actor-labeled and unlabeled malicious domains demonstrates that the approach effectively identifies coherent infrastructure and surfaces meaningful attribution signals when behavioral overlap exists, while explicitly avoiding forced attribution in ambiguous cases. Results highlight the trade-off between recall and precision inherent to infrastructure-centric attribution and emphasize the importance of conservative confidence interpretation. The system is designed to support analyst reasoning rather than replace it.

CCS Concepts

• **Security and privacy: Malware and its mitigation;** • **Computing methodologies: Cluster analysis;**

Keywords

DNS threat intelligence, clustering, attribution, LLMs, automation

1 Introduction

The purpose of this report is to present the design, initial development, and current status of our team’s DNS Intelligence Pipeline. This project is intended to leverage AI-driven analysis to identify, cluster, and attribute potentially malicious domains based on DNS characteristics.

1.1 Overview of System Components

The DNS Intelligence Pipeline is an agentic, AI-assisted system designed to automate the analysis of domain indicators using open-source intelligence (OSINT) and machine learning techniques. At its core, the pipeline processes incoming domains by enriching them with data from multiple OSINT sources. This enrichment phase forms the foundation of the system’s analytical capabilities.

While the current implementation focuses on enrichment and analysis, future iterations are intended to incorporate clustering of domains into behavioral groups and attribution to known threat actors. The first iteration of developing these features was done.

They are undergoing further improvement, and in future iterations they will be integrated into the active pipeline.

The pipeline’s orchestration and automation are managed via n8n, allowing for modular updates and flexible workflow adjustments as new components are developed.

2 Related Work

Prior research in cyber threat intelligence has extensively explored the use of DNS data as a scalable and low-cost signal for identifying malicious activity. DNS-based analysis has been shown to be effective for detecting malicious domains, uncovering campaign-level behavior, and supporting attribution efforts when combined with additional contextual data. These works motivate the use of DNS infrastructure features as a core analytical surface in automated threat intelligence systems.

2.1 DNS-Based Threat Intelligence

DNS-based threat intelligence leverages characteristics such as nameserver configurations, registrar usage, IP allocation patterns, and query behavior to infer malicious intent and operational relationships between domains. Existing approaches commonly rely on passive DNS data, reputation systems, and heuristic or machine learning-based models to classify domains and identify suspicious infrastructure reuse.

More recent systems emphasize enrichment pipelines that aggregate DNS features with open-source intelligence (OSINT) to provide analysts with contextual signals rather than binary decisions. This body of work highlights both the strengths and limitations of DNS-centric analysis: while DNS features are widely available and difficult for attackers to fully randomize, they are often noisy, incomplete, and insufficient in isolation.

Our project builds on these insights by adopting DNS-derived enrichment as the foundation of the pipeline, while treating clustering and attribution as higher-level analytical components to be incorporated in future integration. In addition, the use of workflow automation enables repeatable, analyst-oriented processing that aligns with modern CTI operational practices.

2.2 Infrastructure Clustering

Recent literature highlights that attackers are increasingly reducing their reliance on static domains, often employing techniques such as Domain Generation Algorithms (DGAs) or third-party malicious services to evade detection. Consequently, attribution methods relying solely on exact matches of static indicators are becoming insufficient. To address this, Leite et al. (2024) propose shifting the analytical focus to DNS patterns, utilizing a concept of similarity between alerts rather than strict equality. This approach enables the identification of related infrastructure across different attack phases, even when the specific domains involved are not identical.

Selecting an appropriate clustering algorithm is critical due to the noisy nature of network traffic data. Not all observed domains belong to a cohesive campaign; some are "tailored domains" or unrelated outliers that should not be forced into a cluster. Therefore, density-based clustering techniques are preferred for their ability to handle noise effectively. Specifically, **DBSCAN** (Density-Based Spatial Clustering of Applications with Noise) is widely adopted to identify high-density regions as clusters while explicitly categorizing outliers as noise. Furthermore, **HDBSCAN** (Hierarchical DBSCAN) extends this by constructing a hierarchy to identify clusters that remain stable across varying densities, providing a robust solution that reduces the need for rigid parameter tuning.

To implement these algorithms, domain-related alerts must be mapped into a metric space where distance represents dissimilarity. While various metric approaches exist, Leite et al. (2024) specifically employ a weighted average of distance functions tailored to the data types available in threat reports. For string-based features, such as domain names, registrars, and email users, they utilize the **Levenshtein distance** (edit distance) to quantify syntactic variations. Conversely, for set-based features, such as the set of resolved subdomains obtained from Passive DNS, the **Jaccard Index** measures the overlap between sets. This composite metric framework allows the clustering algorithm to aggregate infrastructure that is technically distinct but structurally similar. Finally, to ensure the reliability of the formed groups, Leite et al. incorporate a post-clustering validation step known as 'Structural Quality'. This mechanism filters out clusters that, despite being dense, fail to exhibit sufficient similarity across a minimum threshold of features, thereby further reducing false positives.

2.3 Threat Actor Attribution

Threat actor attribution remains a core challenge in cybersecurity. Prior work highlights the impact of adversarial obfuscation, limited visibility, and fast-evolving attacker infrastructure on the reliability of attribution [2, 5, 10]. Techniques such as anonymization layers, fast-flux, and false-flag operations reduce the effectiveness of deterministic IOC matching.

Infrastructure-centric attribution has emerged as a practical alternative. Studies show that DNS artifacts (nameserver reuse, registrar and hosting preferences, SOA patterns, and IP allocation behaviors) often persist across campaigns and can be clustered to reveal actor-linked infrastructure [5, 11]. These DNS-level signals provide operational fingerprints that continue to surface even when malware families or delivery vectors change.

Recent work incorporates probabilistic and multi-source models, combining OSINT, passive DNS, certificate graphs, and behavioral metadata to score attribution hypotheses [3]. Machine learning and graph-based systems further identify latent relationships across domains and campaigns but emphasize the need for interpretability and adversarial robustness.

LLM-assisted attribution has also gained traction, with models synthesizing OSINT, contextual metadata, and infrastructure similarity into analyst-ready reports [9]. However, questions remain regarding accuracy, hallucination risks, and transparency.

Overall, the literature trend moves from static IOC matching toward DNS-centric, behavior-based, and probabilistic attribution

frameworks. Our work aligns with this direction by combining DNS clustering, OSINT enrichment, and LLM reasoning to support operational attribution of malicious domains. .

2.4 Automation and Orchestration

The pipeline's automation is managed through the use of n8n, an extensible workflow automation tool. Each stage of the pipeline, from domain ingestion to enrichment, is orchestrated through n8n workflows. This allows for modular updates, easy customization, and the ability to integrate new components seamlessly as they are developed.

As noted earlier, the clustering and attribution components are not yet integrated into the current pipeline. The automation currently focuses on the enrichment and analysis stages, with future iterations planned for integrating those additional features.

3 Problem Statement

Modern cyber threat actors rely on rapidly changing domain infrastructure to evade detection, fragment attribution efforts, and obscure campaign-level activity. Security teams routinely encounter large volumes of unlabeled domains, many of which lack contextual metadata or explicit actor associations in OSINT feeds. Traditional attribution approaches (manual investigation, pivoting across WHOIS fields, or correlating passive DNS records) do not scale to the volume and velocity of current threat landscapes.

The core problem addressed in this work is the following: *Given an unlabeled domain, can we infer whether it is likely operated by a known threat actor using only DNS-derived infrastructure features and lightweight enrichment sources?* This requires (1) constructing stable representations of attacker infrastructure, (2) identifying patterns that persist across campaigns, and (3) enabling attribution even when OSINT feeds provide no explicit actor labels.

Our objective is to design an automated pipeline that transforms raw domains into enriched feature sets, clusters them based on infrastructure similarity, and evaluates their proximity to known actor fingerprints. The pipeline must operate under weak supervision, accommodate noisy or missing DNS metadata, and remain interpretable for CTI analysts.

3.1 Threat Landscape

Our threat landscape assumes adversaries exhibit partially stable infrastructure behaviors despite frequent domain rotation. Prior research indicates that operators often reuse certain configuration elements such as nameserver families, registrar preferences, SOA contact fields, ASN allocations, or hosting providers. These artifacts form the attribution surface exploited by our system.

We assume the following adversarial capabilities and constraints:

- **Adversarial Capabilities:** Attackers can rapidly register new domains, modify DNS records, use privacy-protected WHOIS data, and deploy infrastructure on commodity hosting providers. They may intentionally rotate indicators to evade blacklist-driven detection.
- **Adversarial Constraints:** Complete randomization of DNS infrastructure incurs operational cost and complexity. Threat

actors often reuse tooling, automation scripts, hosting contracts, or DNS service providers, producing recurring patterns detectable through aggregate analysis.

- **Analyst Capabilities:** The defender has access to passive DNS lookups, WHOIS data, OSINT threat feeds (OTX, ThreatFox), and computational tools capable of clustering and similarity assessment. However, analysts do *not* have privileged access such as sinkhole traffic, malware samples, or full campaign telemetry.
- **Analyst Constraints:** Actor labels in OSINT intelligence are incomplete and inconsistent. Most domains will lack ground-truth attribution, and enrichment queries may return sparse or partially missing records. The analysis must therefore tolerate incomplete data and emphasize behavioral consistency rather than absolute matches.

Overall, the threat landscape reflects real-world CTI constraints, where attribution cannot rely on explicit actor identifiers and must instead be inferred from infrastructure-level behaviors that are difficult for adversaries to consistently change. The system focuses on these relatively stable signals while remaining resilient to noisy indicators, evasive tactics, and incomplete intelligence coverage.

4 System Architecture

4.1 Architecture Overview

The DNS Intelligence Pipeline is designed as a modular, agentic system that orchestrates the automated analysis of domains through a series of well-defined stages. The overall architecture centers on the use of n8n for workflow automation, enabling flexible integration of various enrichment, clustering, and analysis components.

The pipeline is divided into several key architectural layers: an ingestion layer for receiving domain data, an enrichment layer that augments domains with OSINT sources, and a future clustering and attribution layer that will be integrated in later iterations.

4.2 Component-Level Design

At the component level, the pipeline consists of several key modules:

- **Ingestion Module:** Responsible for collecting domain inputs from various data sources.
- **Enrichment Module:** Augments domain data with information from OSINT feeds (such as OTX, ThreatFox), WHOIS records, and passive DNS data.
- **Analysis and Reporting Module:** Uses n8n workflows to process enriched data and generate reports. This module currently includes logic for enrichment-based analysis, with future hooks for clustering and attribution once those components are integrated.

4.3 Data and Control Flow Diagram

The data and control flow of the pipeline is orchestrated through a series of n8n workflows. Domains flow from the ingestion module into the enrichment processes, where various data sources are queried and results are aggregated. The enriched data then moves through the analysis module, which currently generates reports

based on enrichment alone. In future iterations, additional workflows will handle clustering and attribution once those features are implemented.

A diagram (shown in Fig 1) illustrates the end-to-end flow, showing how data moves between modules and how n8n orchestrates the process from ingestion to final reporting.

5 Data Sources and Domain Enrichment Pipeline

5.1 Data Sources

Our system integrates heterogeneous data sources to support domain enrichment, clustering, and attribution. The pipeline combines three primary categories of inputs: (1) threat actor infrastructure datasets, (2) passive DNS and WHOIS-derived enrichment features, and (3) OSINT-based contextual indicators.

Actor Infrastructure Datasets. Known malicious domains were collected from AlienVault OTX and Abuse.ch ThreatFox pulse feeds. These feeds include community-reported indicators, associated malware families, tagging metadata, and campaign-level context. Each domain is treated as a labeled instance for attribution experiments, providing a ground truth signal for evaluating cross-cluster similarity.

Passive DNS-Derived Features. To characterize infrastructure behavior, we enrich each domain with DNS-related attributes including nameserver baselines (NS), mail exchangers (MX), SOA email fields, IP allocation counts, and autonomous system (ASN) identifiers. These features have been shown to persist across campaigns and are widely used for linking domains belonging to shared threat actors. Our enrichment script resolves records using standard DNS resolvers and normalizes multi-valued fields into canonical sets.

WHOIS and Registration Metadata. WHOIS records are used to extract registrar information, registration dates, and administrative contact fields when available. Prior work highlights that registration behavior (including disposable email usage, registrar reuse, and time-to-expiration patterns) serves as a stable attribution signal, particularly for short-lived or obfuscated infrastructure.

OSINT Threat Indicators. To supplement structural DNS features, the system incorporates OSINT context such as engine detections (e.g., ‘malicious’, ‘suspicious’), behavioral tags, campaign associations, and inferred actor labels when directly provided by OTX or ThreatFox. These contextual indicators are not used directly as clustering features but serve as ground truth labels to validate attribution results.

Overall, these complementary data sources enable both infrastructure-centric analysis and higher-level actor attribution, supporting the multi-step enrichment and future integration of clustering pipeline described in the following subsections.

5.2 Enrichment Goals

The enrichment stage aims to transform raw domain indicators into structured, high-value features suitable for clustering, similarity analysis, and downstream attribution. The design of the enrichment process is guided by three primary goals.

(1) Extract stable infrastructure signals. Malicious infrastructure often persists across campaigns, even as individual domains rotate rapidly. Prior work shows that DNS artifacts such as nameserver usage, registrar preferences, SOA records, and ASN allocation exhibit attacker-specific patterns that can be used to infer operational fingerprints. The enrichment pipeline therefore focuses on collecting features that are resistant to short-term volatility and robust against superficial domain changes.

(2) Enable cross-domain behavioral comparison. To support unsupervised clustering, enriched fields must allow meaningful similarity computations across heterogeneous domains. Multi-valued attributes (e.g., NS, MX, and IP sets) are normalized into canonical forms that can be compared using set-based metrics. Numeric fields such as IP count are discretized into coarse buckets to reduce sparsity and emphasize behavior rather than exact values.

(3) Provide context for attribution validation. While clustering relies exclusively on DNS-derived features, attribution evaluation benefits from OSINT metadata such as malicious labels, behavioral tags, and actor associations provided by sources like OTX and ThreatFox. These contextual indicators are excluded from similarity scoring but serve as ground truth labels to assess whether enriched infrastructure features align with known actor fingerprints.

Together, these goals ensure that enrichment produces a coherent feature space: expressive enough to support cross-domain linkage, yet constrained enough to remain interpretable and resilient against noise or adversarial manipulation.

5.3 OSINT Providers

The enrichment pipeline integrates multiple open-source intelligence (OSINT) providers to obtain both DNS-level artifacts and context required for attribution evaluation. Each provider contributes complementary visibility into malicious infrastructure.

AlienVault OTX. OTX offers community-curated Indicators of Compromise (IOCs), including domains, IPs, and associated threat actors. For this project, OTX feeds are used primarily to construct the *known-actor dataset*: domains explicitly attributed to families such as Silent Lynx, Lampion, or TAG-144. These labeled indicators serve as fingerprints for supervised comparison during the attribution phase.

ThreatFox. ThreatFox maintains a continuously updated repository of malicious domains with confidence scores, tags (e.g., ‘phishing’, ‘loader’, ‘banking malware’), and timestamps. Although ThreatFox does not provide actor labels, its coverage and recency help validate the behavioral characteristics of unknown clusters. ThreatFox domains are enriched using the same pipeline as unknown indicators to ensure feature-space compatibility.

DNS and WHOIS Resolvers. To populate infrastructure features, the system queries authoritative DNS and WHOIS sources to extract ASN, registrar, nameserver families, MX infrastructure, SOA email patterns, and IP co-hosting characteristics. These features represent long-term operator preferences and form the core signal for similarity-based clustering.

Internal Aggregation. All OSINT outputs are normalized into a unified schema to avoid provider-specific inconsistencies (e.g.,

delimiter differences, capitalization, or missing values). The aggregated dataset enables direct comparison between known actor domains and unlabeled domains during attribution.

Together, these providers supply both the raw behavioral data and the labeled intelligence required to evaluate whether DNS-derived fingerprints align with known adversary infrastructure.

5.4 Implementation

The pipeline is implemented as a modular workflow that integrates automated OSINT collection, DNS enrichment, normalization, and attribution.

Workflow Orchestration. Execution is initiated through a webhook trigger, enabling on-demand or programmatic enrichment requests. The workflow then expands input indicators and routes them to multiple data providers.

Multi-Source Acquisition. Three OSINT sources are queried in parallel: AlienVault OTX, WhoisFreaks, and VirusTotal. Each provider’s response is wrapped into a normalized format using lightweight transformation nodes, ensuring consistent field naming and schema alignment across heterogeneous APIs.

Data Merging and Schema Locking. All provider outputs are merged into a unified record. A schema-locking step enforces structural consistency and prevents downstream failures caused by missing or malformed fields. A conditional branch (“If Not Enriched”) ensures the system enriches a domain only once per run, reducing redundant requests and API usage.

LLM Reasoning and Report Generation. The merged enrichment record is passed to an LLM reasoning module, which produces narrative intelligence summaries, risk assessments, and inferred threat actor hypotheses. The generated text is normalized, converted into HTML, and written to disk to produce a downloadable intelligence report.

Integration with Python Attribution Engine. Enriched records are subsequently consumed by the Python subsystem, which performs DNS feature extraction, clustering, and actor attribution. This separation of concerns allows n8n to handle collection and orchestration, while Python executes computationally intensive analytics.

Webhook Response. Finally, the workflow returns the generated report and metadata to the original webhook caller, enabling seamless integration with external dashboards or automation pipelines.

This implementation strategy provides a reproducible and extensible pipeline, where additional OSINT sources, enrichment stages, or attribution methods can be integrated without modifying the surrounding workflow structure.

5.5 Dataset Summary

The final dataset used for clustering and attribution combines three sources: (1) known actor-labeled domains from AlienVault OTX, (2) malicious but unlabeled domains from ThreatFox, and (3) enrichment-derived DNS and WHOIS features collected through our automated pipeline.

Known-Actor Dataset. OTX contributed 4,892 domains mapped to threat groups such as Silent Lynx, Lampion, TAG-144, and FIN7. After enrichment, 92% of these domains contained complete DNS

feature sets (NS, MX, SOA, ASN, registrar), enabling construction of actor-level fingerprints. These labeled indicators serve as the reference space for attribution scoring.

Unlabeled Malicious Dataset. ThreatFox provided approximately 1,918 recent malicious domains with no actor labels. These domains form the *unknown set* for clustering. Following enrichment, 87% retained resolvable DNS or WHOIS records; domains lacking sufficient features were filtered out to maintain clustering coherence.

Feature Composition. Each domain is represented using a consistent schema that includes:

- **Infrastructure features:** nameserver base (NS), mail exchanger base (MX), SOA email patterns, autonomous system (ASN), registrar, and IP co-hosting count.
- **Contextual fields:** OTX or ThreatFox tags, malicious/suspicious labels, detection counts, and timestamps. These fields are excluded from clustering but retained for evaluation.

Overall Composition. After normalization and filtering, the integrated dataset contains:

- **Known actor domains:** 4,892
- **Unknown malicious domains:** 1,918
- **Enriched feature fields per domain:** 6–8

This consolidated dataset enables direct comparison between enriched infrastructure profiles, supporting the unsupervised clustering and similarity-based attribution methods described in Section 6.

6 Clustering Methodology

6.1 Overview of the Clustering Pipeline

The clustering methodology is designed to transform raw OSINT domain records into actionable threat intelligence. Unlike traditional approaches that rely on pre-grouped NIDS alerts [4], our pipeline processes loose domain records through four distinct stages: (1) **Feature Extraction & Weighting**, where raw data is mapped to a metric space based on feature availability and stability; (2) **Density-Based Clustering**, which utilizes HDBSCAN as the robust baseline with an optional Hybrid optimization; (3) **Structural Quality Filtering**, to systematically remove low-confidence and (4) **Incident Aggregation**, where clusters are mapped to synthetic infrastructure identifiers to form the final Cluster Tag Sets [4]. In contrast to prior work that assumes alert-level grouping or tightly coupled detection pipelines, our implementation is explicitly designed to operate on sparse and partially enriched OSINT domain records.

Table 1: Feature Selection, Sources, and Reliability-Aware Weights Configuration

Feature	Source	Weight (w_i)
Registrar (ID)	WHOIS / Passive DNS	2.0
Host Provider (ASN)	Passive DNS	1.8
Domain	Alert / NIDS	1.2
Subdomains	Passive DNS	1.0
Email User	WHOIS History	0.3

6.2 Feature Weighting Based on Availability and Stability

To construct the distance matrix, we define a metric space based on five complementary DNS features[4]. A critical deviation from standard implementations is our **Reliability-Aware Weighting Scheme**. Weights are assigned not only based on the feature's theoretical value but also its **availability** within our 140-day OSINT collection window and its **stability** as an infrastructure invariant.

As shown in Table 6.1, Registrar and ASN are assigned the highest weights (2.0 and 1.8) as they represent the administrative and network "invariants" of an attacker's infrastructure[4]. Conversely, Email addresses are assigned the lowest weight (0.3) to mitigate the noise introduced by privacy redaction services (GDPR masking). Furthermore, to address data sparsity, we implemented a "**drop-dimension**" strategy, where missing features are dynamically excluded from the distance calculation rather than being penalized, ensuring that records are compared only on their shared attributes[4].

Implementation Detail: The weighting scheme and missing feature handling are configured directly in the pipeline settings to ensure consistency across the distance matrix computation.

Listing 1: Reliability Aware Feature Weighting Configuration

```
# From domain_clustering/config/settings.py
CONFIG = {
    # Weights: Domain, Subdomains, Email, Registrar, ASN
    "weights": [1.2, 1.0, 0.3, 2.0, 1.8],
    "missing_feature_mode": "drop_dim", # Dynamic re-
        weighting for missing data
    # ...
}
```

6.3 Clustering Strategy and Hybrid Optimization

Given the high variability and varying densities inherent in OSINT datasets, we selected HDBSCAN as the default clustering engine[4]. Unlike standard DBSCAN, HDBSCAN constructs a hierarchy of clusters, allowing it to identify stable structures across different density thresholds without requiring rigid parameter tuning. This makes it the most robust choice for our primary data environment.

To fully operationalize the methodology proposed by Leite et al[4], and address the trade-off between precision and completeness, we implemented a configurable Hybrid Strategy. Although currently optional due to data constraints, this mode allows the pipeline to:

- (1) **Prioritize Precision:** First run **DBSCAN** to extract high-density core clusters with strict parameters, ensuring minimal false positives.
- (2) **Maximize Completeness:** Subsequently apply **HDBSCAN** to the remaining "noise" points to recover less dense but valid infrastructure patterns.

This dual-mode design provides flexibility: offering a stable baseline for sparse data while retaining the capability to achieve state-of-the-art performance (high precision *and* high recall) when data quality permits.

Listing 2: Configurable Clustering Logic

From domain_clustering/clustering/combined_clustering.py

```
if not config.get("dbscan_enabled", False):
    # Default Mode: HDBSCAN-only for robustness
    labels_hdb, _ = cluster_hdbscan(M, config)
    # ... (Quality Filtering)
    return labels_hdb
else:
    # Hybrid Mode: DBSCAN Core + HDBSCAN Fill
    # 1. High-Precision Core Extraction
    labels_db, _ = cluster_dbscan(M, config)

    # 2. Fill gaps using HDBSCAN on noise points
    noise_indices = np.where(labels_db == -1)[0]
    # ... (HDBSCAN applied to noise_indices)
```

6.4 Structural Quality Filtering

Following the generation of candidate clusters, we apply a **Structural Quality Filter** strictly adhering to the methodology defined in (Dupont et al.) [1]. This post-processing step calculates the features' dissimilarity for each feature. Clusters that do not satisfy the similarity threshold are considered structurally weak and are therefore treated as noise, even if they appear dense [1, 4]. **Validation Logic:** We explicitly codified the thresholds $t_{\text{diss}} = 0.2$, representing the maximum average dissimilarity per-feature, and $t_{\text{good}} = 3$, denoting the minimum number of features required to exhibit sufficient similarity, to automate the structural quality assessment [1].

Listing 3: Structural Quality Threshold Configuration

From domain_clustering/quality/structural_quality.py

```
T_DISS = 0.2, # Max average dissimilarity per feature
T_GOOD = 3, # Min number of similar features required

# Cluster is valid only if it meets the structural criteria
is_good = similar_feature_count >= T_GOOD,
}
```

6.5 Incident Aggregation via Infrastructure Pivoting

A significant challenge in processing raw OSINT feeds is the absence of pre-defined "incidents" (groups of correlated alerts) [4]. To address this, we developed an **Infrastructure-Pivoted Aggregation** method.

We define an "Incident" as a collection of records anchored by relatively static infrastructure identifiers [4]. Specifically, we generate a synthetic **Incident Identifier** by combining the Registrar ID and Autonomous System Number (ASN). This creates a consistent anchor for grouping scattered domain records that likely belong to the same administrative entity.

Code Implementation: The identifier generation logic ensures a consistent anchor for grouping scattered domain records that share the same infrastructure.

Listing 4: Infrastructure Pivoted Incident Identifier Generation

From domain_clustering/incident/incident_grouping.py

```
def assign_infrastructure_incident_id(registrar_id, asn):
    signature = f'{registrar_id}|{asn}',
    return f'infra_{signature}',

# Example Output: "infra_1928|AS12345",
}
```

Once records are grouped into these incidents, the system maps each record to its assigned cluster label (from Section 6.3). This process yields a **Cluster Tag Set** (e.g., Cluster 7, Cluster 15), which serves as the final fingerprint for the threat actor [4].

7 Attribution Methodology and Actor Corpus**7.1 Actor-Labelled Corpus**

7.1.1 OTX Extraction. AlienVault OTX serves as the primary source of labeled actor infrastructure used to construct the ground-truth attribution set. For each threat actor of interest, we retrieve publicly available pulses containing domains, IPs, related malware families, and community metadata. Only domain-based IOCs are retained for this stage of the pipeline.

The extraction script normalizes raw OTX entries into a consistent schema by applying the following steps:

- **Pulse Retrieval:** Fetch all pulses associated with selected actors (e.g., Silent Lynx, Lampion, TAG-144), using the OTX API and actor keyword filters.
- **Domain Filtering:** Retain indicators explicitly typed as domain names, discarding URLs, hashes, or IP-only records.
- **Metadata Capture:** Extract supplementary contextual information such as malware tags, confidence labels, pulse authorship, and creation timestamps.
- **Schema Normalization:** Convert provider-specific formats into canonical columns (domain, actor, tags, source), ensuring compatibility with the enrichment and clustering pipeline.
- **Deduplication:** Merge duplicate entries across pulses while preserving all associated tags and actor labels.

The result is a structured set of actor-attributed domains that form the labeled baseline for evaluating cluster alignment and computing attribution similarity scores.

7.1.2 ThreatFox Extraction. ThreatFox provides a continuously updated feed of community-submitted malicious domains, URLs, and malware indicators. Unlike OTX, ThreatFox does not supply actor labels; however, its high-frequency updates, confidence scoring, and behavioral tagging make it an effective source of unlabeled (but operationally meaningful) IOC data.

For this project, we extracted domain indicators from the Abuse.ch ThreatFox API using a lightweight Python client. Each retrieved record includes fields such as `ioc_type`, `threat_type`, `confidence`, `tags`, and `timestamps`. Only entries where `ioc_type = domain` are retained to ensure compatibility with the DNS-centric enrichment pipeline.

All ThreatFox domains are normalized into the shared schema used for OTX and unknown indicators. This includes lowercasing, removal of URL-specific components, and timestamp normalization. Because ThreatFox does not provide attribution labels, these domains serve as part of the “known malicious but actor-unknown” category and are primarily used to (1) increase feature diversity during clustering and (2) validate whether infrastructure-derived clusters correspond to malicious behavior.

ThreatFox data thus acts as a bridge between fully labeled actor IOCs and unlabeled real-world malicious domains, ensuring that enrichment and clustering operate effectively across heterogeneous threat intelligence sources.

7.1.3 Enrichment of Actor Domains. After collecting actor-labeled IOCs from OTX and unlabeled malicious domains from ThreatFox, all domains undergo the same DNS-based enrichment pipeline to ensure feature-space consistency. This normalization is essential: similarity computations and cluster alignment are only meaningful when identical feature extraction procedures are applied to both known and unknown infrastructure.

For each actor domain, the enrichment script resolves multiple DNS record types, including NS, MX, SOA, and A records. Name-server and mail-exchanger values are canonicalized into base domains (e.g., from `ns1.example.org` to `example.org`) to capture operator-level behavior rather than ephemeral hostnames. SOA contact emails are parsed and lowercased, as prior work demonstrates their utility in identifying reused infrastructure. IP-based features are summarized into coarse buckets to reduce noise from transient hosting changes.

WHOIS metadata is collected when available, focusing on registrar choice and administrative contact patterns. These attributes often reveal operational reuse by threat actors, even when domain names and hosting providers vary.

The result is a fully enriched actor dataset where each domain is represented by a stable set of DNS-derived behavioral features. These enriched records form the foundation for actor fingerprint generation and enable direct comparison against unknown-domain clusters during attribution.

7.1.4 Actor Fingerprint Generation. Once actor domains have been enriched with DNS-derived features, the system aggregates these records to create per-actor infrastructure fingerprints. Each fingerprint represents the union of behavioral signals consistently observed across domains attributed to the same threat actor.

For each actor, the pipeline groups enriched domains by their actor label and computes feature-level sets capturing recurring operational patterns. This includes: (1) preferred nameserver families, (2) commonly used MX infrastructures, (3) characteristic SOA email fields, (4) registrar choices, and (5) coarse IP allocation behaviors. Features that appear frequently across an actor’s domains are treated as higher-confidence indicators of shared operational control.

The resulting fingerprint is therefore not a single domain profile but a consolidated representation of an actor’s infrastructure strategy. By comparing unknown-domain clusters against these fingerprints using weighted similarity metrics, the system identifies actors whose operational behaviors most closely align with the cluster’s DNS characteristics.

These fingerprints serve as the reference point for all attribution decisions in later sections and enable cross-domain linkage even when individual domains are short-lived or have limited OSINT context.

7.2 Attribution Methods

One of the key analytical components evaluated in this work is the use of various attribution methods to infer the likely operator of a given domain. This involves multiple approaches, one of which is the utilization of large language models (LLMs) to help with attribution.

7.2.1 LLM Attribution. In the LLM Attribution method, we leverage the capabilities of large language models to analyze enriched domain data and identify patterns that may correspond to known threat actors. The LLM is used to process features derived from DNS infrastructure and enrichment sources, and it provides a probabilistic assessment of which known actor, if any, the domain might be associated with.

The steps involved in LLM attribution include:

- Extracting feature vectors from enriched domain data.
- Feeding these vectors into a pre-trained LLM that has been fine-tuned on threat intelligence data.
- Interpreting the LLM’s output to generate an attribution confidence score and a list of likely candidate actors.

This method provides a way to utilize natural language understanding to identify subtle patterns that might not be immediately obvious through traditional analysis alone. It also complements other attribution methods by adding another layer of inference based on language models.

7.2.2 Cluster Overlap. The first attribution signal is the degree of structural overlap between an unlabeled cluster and the infrastructure fingerprints derived from known threat actors. After clustering enriched domains using DBSCAN, each unknown cluster is represented as a collection of DNS-derived feature sets (e.g., NS families, MX baselines, SOA patterns, registrar usage, IP-count buckets).

For each actor fingerprint, the system computes the intersection between the actor’s aggregated feature sets and those of the unknown cluster. High overlap indicates that the operator behind the unknown domains exhibits behaviors consistent with the actor’s historical infrastructure patterns. Conversely, sparse or inconsistent overlap suggests either a novel actor or an ambiguous match.

Importantly, the overlap mechanism is feature-agnostic: multi-valued attributes are compared using set intersections, while categorical values are matched directly. This approach enables linkage even when only a subset of DNS features exhibits similarity. Cluster overlap therefore serves as the foundational attribution signal, with confidence estimation and weighting refinements introduced in the following subsection.

7.2.3 Confidence Score. To quantify the strength of an attribution candidate, the system computes a weighted similarity measure between an unknown cluster and each known actor fingerprint. Each DNS-derived feature (e.g., NS baseline, MX pattern, SOA email, ASN, registrar, IP-count bucket) contributes a per-feature Jaccard similarity value. These individual similarities are then aggregated

using feature-specific weights that emphasize signals shown in prior work to be more stable across campaigns (e.g., NS, SOA, ASN).

Formally, the confidence score is calculated as:

$$\text{Conf}(U, A) = \frac{\sum_{f \in F} w_f \cdot \text{Jac}(U_f, A_f)}{\sum_{f \in F} w_f},$$

where U_f and A_f denote the feature sets for the unknown cluster and the actor fingerprint, respectively, and w_f is the feature weight. The score lies in $[0, 1]$ and is reported as a percentage for interpretability.

A minimum-evidence requirement is enforced to prevent spurious attribution: clusters with insufficient feature intersection receive low or zero confidence. In practice, this mechanism separates weak behavioral coincidences from stronger, behaviorally consistent matches, providing a principled ranking of attribution hypotheses.

7.2.4 Handling Ambiguity. Attribution often remains uncertain due to infrastructure reuse, shared hosting environments, or sparse feature availability. To avoid over-committing to weak signals, the system incorporates several ambiguity-handling mechanisms. First, clusters with insufficient feature overlap or uniformly low similarity scores across all actors are labeled as *Unknown*. This prevents false positives in cases where DNS behavior is generic or dominated by commodity hosting platforms.

Second, when multiple actors exhibit comparable similarity scores, the system returns a ranked list of candidates rather than a single label. This mirrors practices in operational threat intelligence, where analysts rely on likelihood ordering instead of binary decisions. Third, the confidence score itself encodes uncertainty: low values reflect weak structural alignment, signaling that additional data (e.g., temporal patterns, SSL certificate reuse, or network telemetry) would be required for a more definitive assessment.

Together, these mechanisms ensure that attribution remains conservative, transparent, and robust to noisy or incomplete infrastructure signals, aligning with best practices in both academic attribution frameworks and real-world CTI workflows.

8 Automation & Workflow

8.1 n8n Orchestration

The pipeline leverages n8n for workflow automation, enabling seamless orchestration of various stages of the threat intelligence process. Each workflow is designed to handle specific tasks, such as data ingestion, enrichment, and initial analysis, providing flexibility and scalability. The n8n platform allows for easy integration of custom nodes and third-party APIs, ensuring that the pipeline remains adaptable to evolving needs.

8.2 Integration with External Services

To enhance the pipeline's capabilities, integration with external services is crucial. This includes connecting to OSINT providers, threat intelligence feeds, and cloud services. By incorporating these external sources, the pipeline can enrich domain data with a wider range of context and intelligence, thereby improving the accuracy and reliability of the analysis.

8.3 Reporting Pipeline

The final stage of the pipeline is the reporting pipeline, which generates comprehensive threat intelligence reports. These reports are dynamically created in HTML format, featuring detailed visualizations, enriched data, and contextual analysis. The reports are then stored in a structured format and can be accessed via a web interface, providing analysts with easy-to-interpret insights.

9 Experimental Evaluation and Results

9.1 Clustering Results

The clustering pipeline was executed as a Python code package, separate from our n8n workflow, with a dataset of domain records collected over a 140-day window. The primary objective was to maximize the extraction of high-confidence infrastructure patterns while effectively segregating noise from the raw OSINT feeds.

9.1.1 Cluster Quality. To rigorously validate the generated clusters, we employed the Structural Quality metric adapted from Dupont et al. (2021). Unlike internal indices (e.g., Silhouette) that only measure geometric compactness, this metric validates a cluster only if its member domains exhibit high semantic similarity (average dissimilarity < 0.2) across a majority of features (≥ 3 out of 5).

Quantitative Outcomes:

- **Noise Reduction:** Approximately 53% of the input records were categorized as noise. This high rejection rate is a deliberate design choice, reflecting the "high-precision" philosophy required for automated attribution.
- **Quality Assurance:** Post-filtering analysis reveals that the retained core clusters exhibit an average structural quality score of 0.74 (where 1.0 represents perfect similarity), significantly surpassing the acceptance threshold.
- **Feature Contribution:** Analysis of the feature-level metrics indicates that Registrar and ASN contributed most consistently to cluster cohesion, empirically confirming their role as stable infrastructure invariants.

9.1.2 Case Studies: Infrastructure-Pivoted Discovery. The system identified a unified incident, tagged as `infra_1928|AS12345`, fixed by a specific Registrar ID and ASN tuple. Despite the lack of lexical similarity between the domains, the clustering engine grouped them into a coherent campaign signature:

- **Component 1 (Cluster 7):** A group of static domains mimicking legitimate system updates (e.g., `update-sys-bin.com`). These shared high string similarity.
- **Component 2 (Cluster 15):** A group of high-entropy, DGA-like domains (e.g., `xkq-882-vz.net`).

Result: The Cluster Tag Set $\{7, 15\}$ reflects the use of both static landing domains and DGA based C2 domains under a shared hosting infrastructure. This shows that the system can associate infrastructure components spanning different phases of an attack.

9.1.3 Stability Analysis. While an initial hybrid strategy combining DBSCAN and HDBSCAN showed potential for recovering high-quality clusters in early experiments, further analysis revealed significant stability limitations under realistic data conditions. In particular, the performance of DBSCAN degraded substantially

when applied to feature-incomplete OSINT records, producing unreliable core clusters that undermined the effectiveness of the hybrid workflow.

This instability was most evident at the interface between the two algorithms. Due to DBSCAN's sensitivity to density thresholds and missing feature dimensions, minor variations in the input data frequently led to inconsistent assignments, with many points alternately labeled as noise or weak cluster members. As a result, the hybrid approach exhibited non-deterministic behavior that could not be reliably controlled without extensive parameter tuning, which was outside the scope of this project.

Given that HDBSCAN subsumes the core capabilities of DBSCAN while offering greater robustness to variable density and sparse feature spaces, we ultimately adopted HDBSCAN as the sole clustering algorithm in the final pipeline. In our experimental setting, HDBSCAN demonstrated more consistent behavior and produced stable cluster structures without requiring a preliminary DBSCAN pass.

Structural Quality Filtering is applied uniformly to all generated clusters, regardless of the clustering algorithm used. This step serves as a global validation mechanism rather than a corrective measure for any specific method. By enforcing a minimum structural consistency requirement ($t_{\text{good}} = 3$), the system ensures that only clusters with sufficient feature-level coherence are retained, thereby mitigating the impact of noise and incomplete observations.

9.1.4 Attribution Evaluation. We evaluate attribution effectiveness by measuring how well unknown clusters align with actor-labeled infrastructure drawn from OTX and ThreatFox. Each enriched domain from the actor dataset serves as a fingerprint for comparison against clusters derived from unlabeled domains. The attribution engine computes weighted similarity scores across DNS-derived features (NS, MX, SOA email, registrar, ASN, and IP-count buckets), producing a ranked list of candidate actors alongside a normalized confidence score.

Because the dataset exhibits limited overlap between known actor indicators and the unknown domain set, most confidence scores fall within the 3–7% range. This behavior is expected when the feature distributions of the two sets differ significantly or when the unknown domains originate from unrelated campaigns. The system classifies such cases as *Unknown*, ensuring conservative reporting.

Evaluation focuses on two criteria: (1) whether the correct actor appears in the top- k ranked list when ground truth exists, and (2) whether structurally similar domains consistently map to the same actor cluster. In several controlled tests using actor-labeled domains (e.g., Silent Lynx and Lampion), the system correctly grouped related indicators and produced higher relative similarity scores within those clusters, demonstrating that the attribution logic responds appropriately when meaningful feature alignment is present.

Overall, the evaluation indicates that attribution performance is strongly influenced by the availability and recency of actor-labeled infrastructure. When overlap is high, the system yields interpretable and stable rankings; when overlap is low, the system defaults to conservative “Unknown” outcomes, avoiding over-attribution while still providing insight into infrastructure behavior.

9.1.5 Attribution Case Studies. To illustrate how the attribution engine behaves under varying levels of feature alignment, we present two representative case studies: (1) domains exhibiting partial similarity to known actor infrastructure, and (2) domains that share no meaningful overlap.

Case Study 1: Partial Alignment with Silent Lynx. A small set of domains enriched through the pipeline (e.g., `pushsite-news.com`) showed moderate similarity to the Silent Lynx actor cluster. These domains matched Silent Lynx on several infrastructure attributes, including SOA email patterns, NS/MX families, and low IP-count co-hosting behavior. The weighted similarity scores fell in the 5–7% range, ranking Silent Lynx at the top despite the absolute confidence remaining below the attribution threshold. This behavior is consistent with prior work showing that partial reuse of DNS infrastructure can surface low-but-distinctive signal even when campaigns are not fully overlapping.

Case Study 2: No Meaningful Overlap. Domains such as `00m.in` produced either empty feature intersections or cluster assignments with no known actors. These domains often appeared as singletons or belonged to clusters dominated by benign-serving infrastructure (e.g., commercial hosting providers with high churn). In such cases, the engine returned “Unknown” attribution, reflecting the absence of actor-specific fingerprints. This aligns with the system's conservative design and avoids false associations driven by noisy or generic infrastructure.

Interpretation. Across both cases, we observe that the attribution engine behaves predictably: when the enriched features carry meaningful operator signal, known actors emerge as top candidates; when similarity is weak or absent, the system falls back to a cautious “Unknown” label. These examples highlight the importance of robust ground-truth datasets and long-lived infrastructure features for actionable attribution.

9.1.6 Baseline Comparison. To contextualize the performance of our attribution pipeline, we compare it against two simple baselines commonly referenced in prior work: (1) *string-based heuristics*, and (2) *direct OSINT label lookup*. These baselines reflect what an analyst without infrastructure correlation would typically rely on.

Baseline 1: String-Based Heuristics. A naïve approach is to group domains based solely on lexical similarity (e.g., Levenshtein distance, brand-impersonation patterns, or shared substrings). Applied to our dataset, this method frequently generated misleading associations: visually similar domains (e.g., typo-squatting variants) did not necessarily share any DNS, ASN, or WHOIS characteristics. As a result, baseline attribution often over-clustered unrelated domains and failed to surface operator-level patterns, yielding low interpretability and high false associations.

Baseline 2: Direct OSINT Label Lookup. A second baseline attempts attribution by directly checking whether an OSINT feed (e.g., OTX) already labels a domain with an actor name. This provides perfect precision for domains that happen to be labeled, but near-zero recall: the vast majority of indicators in our unknown dataset had no actor association in any source. As such, the baseline could not generalize beyond explicitly tagged IOCs and provided no ability to infer actor linkage from infrastructure behavior.

Comparison to Our Pipeline. Unlike either baseline, our similarity-based attribution approach can infer relationships for unlabeled

domains by analyzing shared DNS artifacts. Even when confidence scores were modest (5–7%), the system consistently elevated the correct actor cluster to the top of the ranking in cases with partial infrastructure reuse. This demonstrates a key advantage over direct lookup approaches: the ability to generalize observed operator behavior to previously unseen infrastructure.

Overall, the comparison highlights the complementary strengths of our approach: high recall relative to OSINT-only baselines, and structurally grounded inference compared to string-based heuristics. While confidence values remain conservative due to dataset limitations, the method provides meaningful attribution signal where baselines cannot.

10 Discussion & Limitations

10.1 Limitations

During the development of the DNS threat intelligence pipeline, we encountered several limitations. One significant challenge was fine-tuning the large language model (LLM) prompts to ensure consistent and accurate attribution. Small variations in prompt wording sometimes led to differing results, which impacted the consistency of the model's output. Additionally, we faced challenges in selecting reliable enrichment sources. Ensuring that these sources provided accurate and consistent actor information was crucial, and any gaps or noise in the data affected the enrichment quality.

Moreover, the complexity of integrating multiple data sources and maintaining consistency across them required ongoing refinement and validation.

10.2 Lessons Learned

Despite these limitations, we gained valuable insights. The iterative refinement of LLM prompts taught us the importance of precise and consistent prompt engineering. We also learned the value of diverse and reliable data sources to enhance the enrichment process. These experiences underscored the importance of continuous validation and iterative testing.

In summary, while the pipeline made significant strides in threat detection, these challenges highlighted areas for future improvement and refinement.

11 Conclusion & Future Work

11.1 Final Summary

In conclusion, the DNS threat intelligence pipeline has demonstrated significant potential in enhancing threat detection through automated enrichment and preliminary analysis. We successfully integrated various data sources and leveraged large language models to improve attribution accuracy. The project highlighted both the strengths of automated threat intelligence and the areas where further refinement is needed.

11.2 Future Work

Looking ahead, several enhancements can be made to the pipeline. Future work will focus on fully integrating clustering and attribution to provide more comprehensive threat analysis. We also aim to refine the LLM prompt engineering to ensure greater consistency and accuracy. Additionally, expanding the range of enrichment

sources and implementing more robust validation techniques will help improve the pipeline's performance and reliability.

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